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PUBLIC SUMMARY

# System 6.9

Community DIY endless shuttle powered by battery or hard-wired. Open software. Big-box builds. Size the motor and pack for your use case.

## Mission

System 6.9 is a **community project**: open-source software plus builds from materials you can get online or at Home Depot, Harbor Freight, Lowe's, Menards, and similar. The center of gravity is community and community development — not a finished commercial tow product.

Each builder has their own use case. Motor size and battery pack (or hard-wired supply) are chosen by that use case — we do not ship a one-size powertrain.

## One-liner

DIY endless shuttle for the community — one architecture, many surfaces. Water (foil/skim), backyard snow, sled uphill, skate tow, and more. You size the motor and pack for your use. We publish the guide, the software and real build costs, all free. Use at your own risk.

## Why build one

Commercial endless tows such as System 2.0 cost tens of thousands of dollars. System 6.9 is for hobbyists who want to compare ideas, understand what a build needs and what it really costs, and then make one for their own lake, pond or hill. Compare community builds or read the free guide.

## Use cases

Same endless loop. Different accessories, mounts, and speed presets.

| USE | SETUP NOTES | HEIGHT / SPEED BIAS |
|---|---|---|
| Foil / skim (water) | Shore A-frames; floating Dyneema | Low tow ~0.8–1.5 m · foil soft / skim faster |
| Backyard snowboard | Hill or flat-to-slope chord; return tow to top | Low–mid · soft start |

| | | |
|---|---|---|
| Uphill sled tow | Steeper chord; stronger tension; kids/adults | Low handle; slow presets; EOR critical |
| Skateboard tow | Parking lot / empty street / private pavement | Low tow; soft ramp; short span OK |
| Wakeskate / wakesurf-style | Same water frames | Low |
| Grass / dirt “summer sled” | Dry hillside practice | Low; expect abrasion on line |
| Ski / snowboard park return | Side hit or backyard feature return | Same as snow uphill |
| Training / rehab pull | Very low speed presets | Softest ramp |
| Cargo / gear shuttle | Non-human: coolers, tools across a cut | Own-risk; different carrier |
| Photo / camera dolly | Slow constant speed along a lane | Smooth speed mode |

Foil / skim · Snow · Sled uphill · Skate tow · Training · Cargo

> Weak fits (don’t lead with these): high big-air wake (wants tall towers), full-size cable parks, public ski-area lifts (certification world).

### Nothing is for sale

Nothing is on sale right now. The guides, the build gallery and the software are free, and the guides will always be free. See the roadmap below for what might come later.

### What you source

- Motors (e.g. BLDC 80100-class)
- ESCs (e.g. VESC-class)
- Batteries and chargers (e.g. 12–14S high-C packs)
- Remotes (community-supported options)
- Scaffold, pipe, Unistrut, lumber and common hardware

You size these for your own use case. The parts list shows what I'm using.

### Software

- Shuttle firmware — open source, free to download and flash from GitHub
- Docs + CAD — open (license finalize at repo launch)
- Community — issues and PRs improve soft start, turn points, end-of-run, dual-site presets

- Warranty on code — **none**. Use at your own risk. Not a certified tow system.

## Architecture

One motor · endless loop · reverse = return. Drive A-frame holds the control box and user powertrain; return A-frame holds the idler and tension.

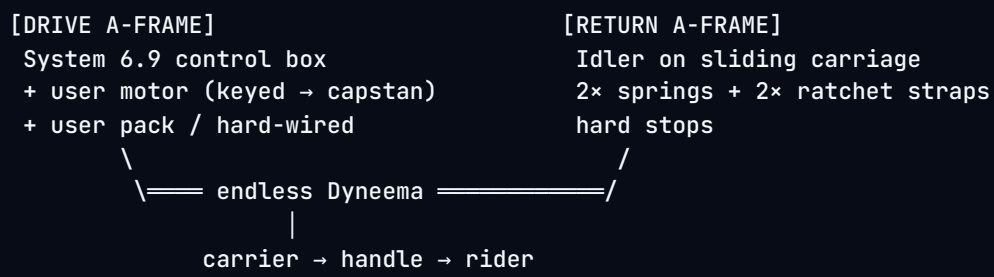
**Drive A-frame**

System 6.9 control box
+ user motor → capstan
+ user pack (or hard-wired)

ENDLESS DYNEEMA
carrier → handle → rider

**Return A-frame**

Idler on sliding carriage
2× springs + 2× ratchet straps
hard stops




## Prototype build

### Phase 0 — Design freeze DONE

One-motor endless loop. Drive A-frame = power + control. Return A-frame = idler + tension. Soft ramp, E-stop, low tow height (~0.8–1.5 m).

### Phase 1 — Bench prove NOW

Keyed hub → capstan grips multi-wrap under a hard dead pull; E-stop kills power. **Do not go to the lake until this passes.**

### Phase 2 — Empty lake shuttle

Full line (~80–150 m span), two A-frames, springs + straps, anchors/guys. Pass = clean reverse cycles with no rider.

### Phase 3 — First rides

Carrier + handle. Foil first (~12–18 km/h), then skim (~25–35 km/h) with spotter + E-stop ready.

## Community roadmap

### 1. Compare builds NOW

The build gallery is open. Post your setup and see how other people solved the same problems.

### 2. Publish real costs

What each build actually cost, part by part, next to the estimates below.

### 3. Get working builds

Builds that pass the test ladder. Their firmware, docs, CAD and parts lists go up on GitHub for anyone to copy.

### Maybe later

Only after there are working builds: affiliate links to parts, merch, custom parts if a build really needs one, and pre-spliced Dyneema loops cut to your length with a tow-link loop for the handle. None of these are on sale now.

> Community DIY. Use at your own risk. Not a certified tow system. Contribute via the home splash signup.

## Parts list

Approximate USD from research spot-checks — will move; use at your own risk. Not a certified tow system. Span ~80–150 m → endless loop ~200–350 m of line. Pick **battery or hard-wired** — don't double-count.

> **Capstan rule:** do not surge bare AmSteel-Blue on aluminum. Use jacketed UHMWPE, polyester drive wraps, or a grooved keyed drum.

### Powertrain — pick one path

| PATH | WHAT'S IN IT | APPROX USD |
|---|---|---|
| **Battery lean** | ~1.5–2 kW BLDC + ESC, 48 V LiFePO4 ~40–50 Ah, light reduction | $560–1,100 |
| **Battery comfortable / adult deep-water skim** | High-kW / 80100-class path + high-C pack + proper reduction | $1,500–4,000+ |
| **Hard-wired lean** | 2–3 HP VFD + 1.5–2 HP keyed 3ϕ + mount/coupling | $450–850 |
| **Hard-wired adult deep-water** | 5–10 HP motor + matching VFD | $600–1,500+ |

> Lean is cruise / youth / assisted-plane. Adult ~100 kg deep-water skim needs ~8–12+ kW peak / ≥~200 kgf — size for the *start*, not cruise. Youth ~30 kg can stay lean–comfortable with torque gearing + a long soft ramp.

## Shared hardware

| BLOCK | NOTES | APPROX USD |
|---|---|---|
| Capstan / keyed drum | Portable-winch-style or DIY grooved keyed drum | $85–400 |
| Return + fairlead sheaves | 2–4 snatch blocks / marine sheaves | $50–250 |
| Keyed hub / bushing | Match motor/gearbox shaft | $40–150 |
| Line 250–350 m | Jacketed 1/4″ UHMWPE, hybrid, or 8–10 mm coated | $1,000–2,200 |
| Two towers | Scaffold / pipe A-frames / lumber POC (temp) | $150–1,200 |
| Tension kit | Springs, ratchet straps, turnbuckles, take-up | $150–400 |
| Anchors / guys | Earth anchors, ballast, braces | $80–300 |
| E-stop + contactor + enclosure + wire | Hard-wired NC E-stop; sized for peak current | $200–600 (+ charger if needed) |
| Carrier + short tow + handle | Rider interface | $70–200 |

## Roll-up totals (two towers, one endless loop)

| BUILD | APPROX TOTAL |
|---|---|
| Lean DIY battery (generic haul / youth-leaning) | ~$2,000–3,500 |
| Hard-wired Phase-1 | ~$2,200–3,500 |
| Comfortable Phase-1 | ~$3,500–5,500 |
| Adult deep-water skim (~100 kg) | ~$4,500–9,000 battery · ~$4,000–7,500 hard-wired |

## Buy order

1. Pick power path (battery vs hard-wired) and use-case size (youth vs adult deep-water).
2. Buy line (longest lead / largest $).
3. Capstan + return sheaves — prove grip before load.
4. Two short towers + guys/anchors.
5. Tension + take-up.
6. Electrical safety panel.
7. Dry run at 20–40 m before full 80–150 m.

8. Empty shuttle → foil → skim.

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